

ENGG2020 DIGITAL LOGIC AND SYSTEMS

CHAPTER 3: GATE LEVEL MINIMIZATION

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CONTENTS

- Canonical and Standard Forms
- Gate level minimization
- K-map



MINTERM

- Normal form and complement form
 - A binary variable may appear either in its normal form, x , or in its complement form, x'
- Minterm
 - If there are 2 variables, x and y , there are 4 possible combinations: $x'y'$, $x'y$, xy' , xy
 - Each of these 4 AND terms is called a **minterm**, or **standard product**
 - If there are N variables, there are 2^N minterms
 - A symbol for each minterm is denoted as m_j , where the subscript j denotes the decimal equivalent of the binary number of the minterm designated

3

MAXTERM

- Maxterm
 - If there are 2 variables, x and y , there are another 4 possible combinations: $x+y$, $x+y'$, $x'+y$, $x'+y'$
 - Each of these 4 OR terms is called a **maxterm**, or **standard sum**
 - If there are N variables, there are 2^N maxterms
 - A symbol for each maxterm is denoted as M_j , where the subscript j denotes the decimal equivalent of the binary number of the maxterm designated
- For minterm, "1" $\rightarrow$ Normal (True), and "0" $\rightarrow$ Complement (False)
- For maxterm, "0" $\rightarrow$ Normal (True), and "1" $\rightarrow$ Complement (False)
- Therefore, m_j is the complement of M_j

4

MINTERMS AND MAXTERMS

Minterms and Maxterms for Three Binary Variables

<i>x</i>	<i>y</i>	<i>z</i>	Minterms		Maxterms	
			Term	Designation	Term	Designation
0	0	0	$x'y'z'$	$=111=1$ m_0	$x + y + z$	$=0+0+0=0$ M_0
0	0	1	$x'y'z$	m_1	$x + y + z'$	M_1
0	1	0	$x'yz'$	m_2	$x + y' + z$	M_2
0	1	1	$x'yz$	m_3	$x + y' + z'$	M_3
1	0	0	$xy'z'$	m_4	$x' + y + z$	M_4
1	0	1	$xy'z$	m_5	$x' + y + z'$	M_5
1	1	0	xyz'	m_6	$x' + y' + z$	M_6
1	1	1	xyz	m_7	$x' + y' + z'$	M_7

5

EXAMPLE (MINTERMS & MAXTERMS)

- For example, we have the truth table of the following functions f_1

Functions of Three Variables

<i>x</i>	<i>y</i>	<i>z</i>	Function f_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

6

EXAMPLE (MINTERMS & MAXTERMS)

- A Boolean function can be expressed algebraically in terms of either minterms or maxterms

- In term of minterms, we have

- $f_1(x, y, z) = x'y'z + xy'z' + xyz$

- $f_1(x, y, z) = m_1 + m_4 + m_7$

- $f_1(x, y, z) = \sum m(1,4,7)$

- $f_1'(x, y, z) = \sum m(0,2,3,5,6)$

Functions of Three Variables

x	y	z	Function f_1
0	0	0	term 0 0
0	0	1	term 1 1
0	1	0	term 2 0
0	1	1	term 3 0
1	0	0	term 4 1
1	0	1	term 5 0
1	1	0	term 6 0
1	1	1	term 7 1

7

EXAMPLE (MINTERMS & MAXTERMS)

- In term of maxterms, we have

- $f_1(x, y, z) = (x + y + z)(x + y' + z)(x + y' + z')(x' + y + z')(x' + y' + z)$

- $f_1(x, y, z) = M_0 M_2 M_3 M_5 M_6$

- $f_1(x, y, z) = \prod M(0,2,3,5,6)$

- $f_1'(x, y, z) = \prod M(1,4,7)$

Functions of Three Variables

x	y	z	Function f_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

8

EXAMPLE OF MINTERM & MAXTERM

- Given a truth table:

A	B	X
0	0	0
0	1	1
1	0	0
1	1	1

- By using Minterm, 1 = True, 0 = False

- $X = AB' + AB$ sum of product (SOP)

- By using Maxterm, 0 = True, 1 = False

- $X = (\bar{A} + B)(A' + B)$ product of sum (POS)

- If $A=0, B=0, X = (0+0)(1+0) = 0$

- If $A=0, B=1, X = (0+1)(1+1) = 1$

- If $A=1, B=0, X = (1+0)(0+0) = 0$

- If $A=1, B=1, X = (1+1)(0+1) = 1$

9

CANONICAL AND STANDARD FORMS

- Standard **Sum-Of-Products (SOP)** forms

- Equations are written as AND terms summed with OR operators

- $F = xyz + xy' + z'$

- Standard **Product-Of-Sums (POS)** forms

- Equations are written as OR terms multiplied with AND operators

- $F = (x + y + z)(x + y')(z')$

- Canonical forms, e.g.

- Sum of minterms: $F = xyz + xy'z + x'yz'$

- Product of maxterms: $F = (x + y + z)(x + y' + z)(x' + y + z')$

- Mixed forms is neither SOP nor POS, e.g.

- $F = (xy + z)(x + y'z) + (x'y + z')$

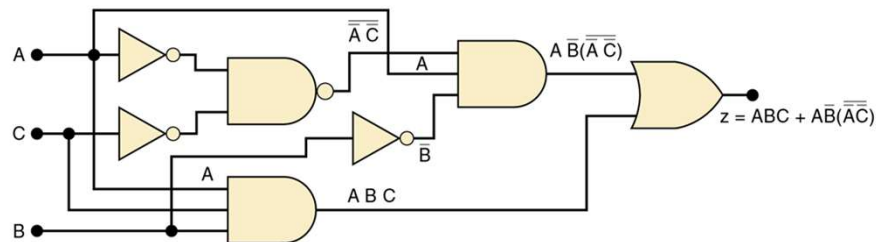
10

11

GATE LEVEL MINIMIZATION

SIMPLIFYING A LOGIC CIRCUIT

- Simplify the logic circuit below:

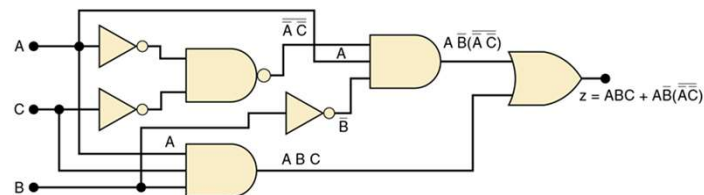


$$\begin{aligned}
 z &= ABC + AB\bar{A}\bar{C} \\
 &= ABC + AB\bar{A}(C + \bar{C}) && \text{[theorem (17)]} \\
 &= ABC + AB\bar{A}C + AB\bar{A}\bar{C} && \text{[cancel double inversions]} \\
 &= ABC + AB\bar{A}C + AB\bar{A}\bar{C} && \text{[multiply out]} \\
 &= ABC + AB\bar{A} + AB\bar{A}\bar{C} && \text{[} A \cdot A = A \text{]}
 \end{aligned}$$

12

SIMPLIFYING A LOGIC CIRCUIT

- Simplify the logic circuit below:

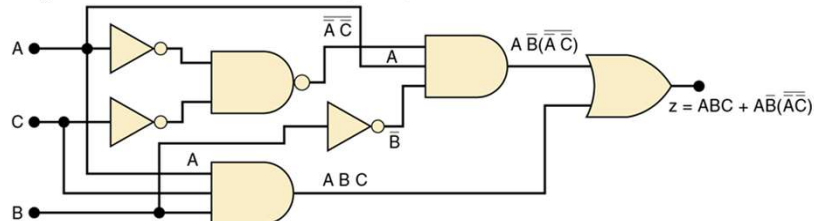


$$\begin{aligned}
 z &= ABC + AB(\overline{\overline{A}} + \overline{\overline{C}}) && \text{[theorem (17)]} \\
 &= ABC + AB(A + C) && \text{[cancel double inversions]} \\
 &= ABC + ABA + AB\overline{C} && \text{[multiply out]} \\
 &= ABC + AB + AB\overline{C} && \text{[} A \cdot A = A \text{]} \\
 z &= AC(B + \overline{B}) + AB \\
 z &= AC(1) + AB \\
 &= AC + AB
 \end{aligned}$$

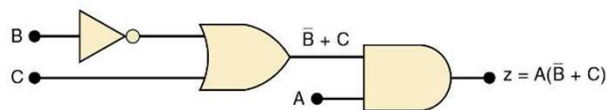
13

SIMPLIFYING A LOGIC CIRCUIT

- Before simplification: $z = ABC + AB'(A'C')'$



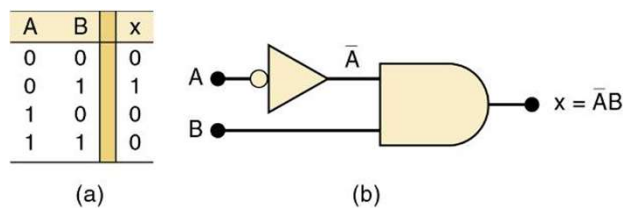
- After simplification: $z = A(C+B')$



14

SIMPLIFYING A LOGIC CIRCUIT

- Step 1: setup its truth table, figure (a)
- Step 2: write an AND term for each row where output is 1
- Step 3: combine the AND terms in SOP form
- Step 4: **simplify the expression by Boolean algebra, if possible**
- Step 5: implement the simplified circuit, figure (b)



15

EXAMPLE (SIMPLIFICATION)

- Simplify a logic circuit with the following truth table

Truth table.

A	B	C	x
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

AND terms for each case where output is 1.

$\rightarrow \bar{A}BC$

$\rightarrow A\bar{B}C$

$\rightarrow AB\bar{C}$

$\rightarrow ABC$

SOP expression for the output:

$$x = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

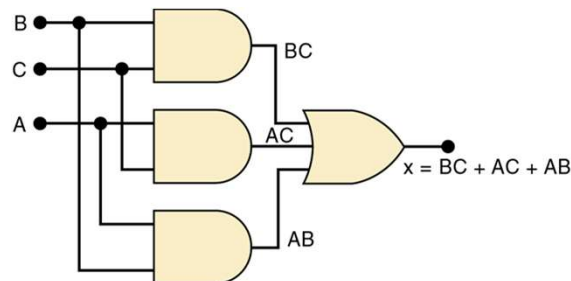
16

EXAMPLE (SIMPLIFICATION)

- Design a logic circuit with three inputs, A, B, and C.

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$x = BC + AC + AB$$



17

SIMPLIFYING A BOOLEAN EXPRESSION

- Simplify the Boolean function, $F(A, B, C) = \sum m(0, 2, 3, 4, 5, 7)$
- Calculation 1:

$$\begin{aligned}
 F &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + ABC \\
 &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + ABC \\
 &= \overline{A}\overline{C}(B + \overline{B}) + BC(A + \overline{A}) + \overline{A}B(\overline{C} + C) \\
 &= \overline{A}\overline{C} + BC + \overline{A}B
 \end{aligned}$$
- Calculation 2:

$$\begin{aligned}
 F &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + ABC \\
 &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + ABC \\
 &= \overline{B}\overline{C}(\overline{A} + A) + \overline{A}B(C + \overline{C}) + AC(B + \overline{B}) \\
 &= \overline{B}\overline{C} + \overline{A}B + AC
 \end{aligned}$$
- Both calculations are correct and having the same numbers of terms, is there any **systematic way** to simplify it?

18

KARNAUGH MAP METHOD

- K-map method is a **graphical method** of simplifying logic equations or truth tables
- Theoretically, it can be used for any number of input variables, but **practically limited** to 5 or 6 variables
- For 2 input variables, x and y

m_0	m_1
m_2	m_3

(a)

		y	
		0	1
x	0	m_0 $x'y'$	m_1 $x'y$
	1	m_2 xy'	m_3 xy

(b)

19

2-VARIABLE K-MAP

- For the following expressions, we have the K-maps below:

- (a) xy
- (b) $x+y$

		y	
		0	1
x	0	m_0	m_1
	1	m_2	m_3

(a) xy

		y	
		0	1
x	0	m_0	m_1
	1	m_2	m_3

(b) $x + y$

$m_1m_3 \rightarrow y$
 $m_2m_3 \rightarrow x$
 so $x + y$
 or
 $m_1m_3 \rightarrow y$
 $m_2 \rightarrow xy'$
 so $xy' + y = y + xy'$
 note: $x+x' = x+y$
 so $y+xy' = y+x = x+y$

20

3-VARIABLE K-MAP

- 3-variable K-map

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

(a)

Only 1 variable changed at the neighborhoods

$x \backslash yz$		y			
		00	01	11	10
x	0	m_0 $x'y'z'$	m_1 $x'y'z$	m_3 $x'yz$	m_2 $x'yz'$
	1	m_4 $xy'z'$	m_5 $xy'z$	m_7 xyz	m_6 xyz'

z

(b)

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21

EXAMPLE (3-VARIABLE K-MAP)

- Simplify $F(x, y, z) = \sum(2, 3, 4, 5)$

$x \backslash yz$		y			
		00	01	11	10
x	0	m_0	m_1	m_3 1	m_2 1
	1	m_4 1	m_5 1	m_7	m_6

z

Annotations: $x'y$ points to m_2 , xy' points to m_4 .

- Ans: $F(x, y, z) = x'y + xy'$

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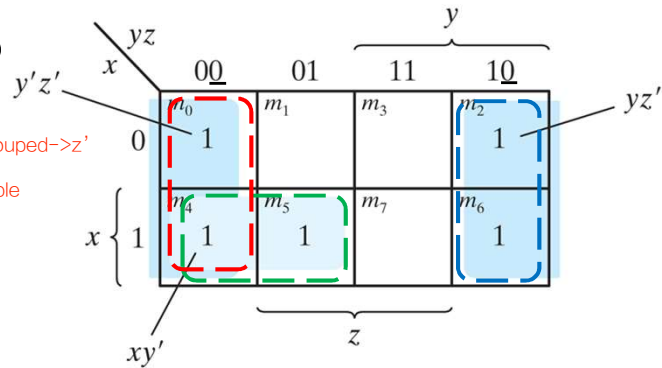
22

EXAMPLE (3-VARIABLE K-MAP)

- Simplify $F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$

red and blue region can be grouped $\rightarrow z'$

larger group, term is more simple
so $m_4 m_5$ group $\rightarrow xy'$



- Ans: $F(x, y, z) = y'z' + yz' + xy' = z' + xy'$

Note: $y'z' + yz' = z'$

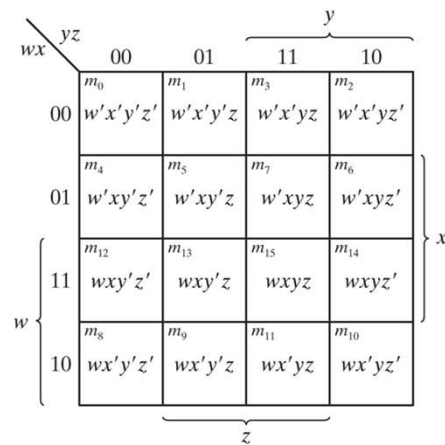
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23

4-VARIABLE K-MAP

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6
m_{12}	m_{13}	m_{15}	m_{14}
m_8	m_9	m_{11}	m_{10}

(a)



(b)

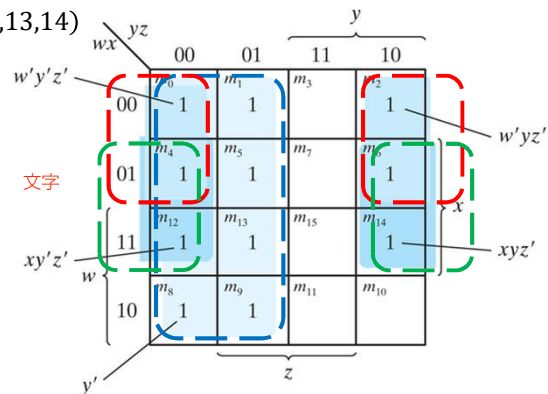
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24

EXAMPLE (4-VARIABLE K-MAP)

- Simplify $F(w, x, y, z) = \sum(0,1,2,4,5,6,8,9,12,13,14)$

recall: group term in 2^n times
 $n=0,1,2$



- Ans: $F(x, y, z) = y' + w'z' + xz'$
 SOP

Note: $w'y'z' + w'yz' = w'z'$
 $xy'z' + xyz' = xz'$

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25

EXAMPLE (4-VARIABLE K-MAP TO POS)

- Simplify $F(A, B, C, D) = \sum(0,1,2,5,8,9,10)$ in POS form

- $F'(A, B, C, D) = \sum(3,4,6,7,11,12,13,14,15)$

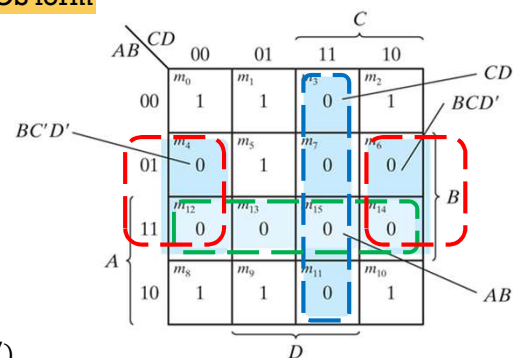
- $F'(A, B, C, D) = BD' + CD + AB$

De Morgan thm

- $F(A, B, C, D) = (BD' + CD + AB)'$

- $F(A, B, C, D) = (BD')'(CD)'(AB)'$

- Ans: $F(A, B, C, D) = (B' + D)(C' + D')(A' + B')$



Note: $BC'D' + BCD' = BD'$

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26

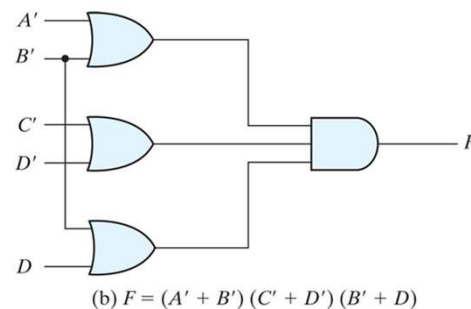
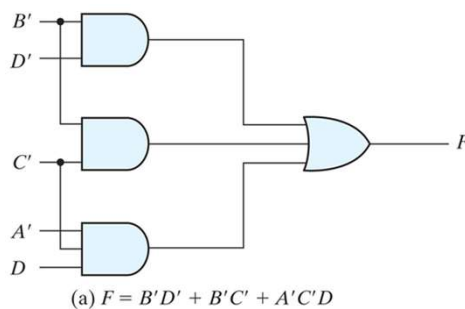
TWO-LEVEL IMPLEMENTATIONS

- From the previous example, $F(A, B, C, D) = \sum(0, 1, 2, 5, 8, 9, 10)$ can be simplified
 - In SOP form by collecting the 1s as $F(A, B, C, D) = B'D' + B'C' + A'C'D$
 - In POS form by collecting the 0s as $F(A, B, C, D) = (B' + D)(C' + D')(A' + B')$
- The implementation of a function in a standard form is said to be a two-level implementation
 - In SOP form, AND gates are connected to a single OR gate
 - In POS form, OR gates are connected to a single AND gate

27

TWO-LEVEL IMPLEMENTATIONS

- (a) In SOP form, AND gates are connected to a single OR gate
- (b) In POS form, OR gates are connected to a single AND gate

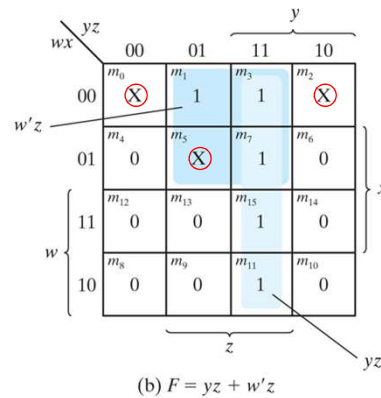
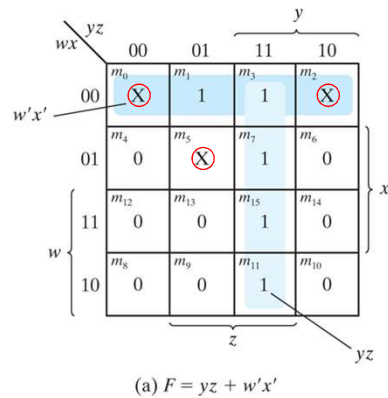


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28

EXAMPLE (K-MAP WITH “DON’T CARE”)

- Simplify $F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$ with m_0 , m_2 , and m_5 are “Don’t Care”.



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29

GUIDELINES OF K-MAP

- Group number of cells in powers of 2
- Group as many cells as possible
- The larger the group, the less number of variables in that term
- Make as few groups as possible to reduce the number of terms

30

EXAMPLE (ALARM DESIGN)

- A manufacturing plant is going to setup an alarm system to monitor a chemical production process
- Sensor inputs include Temperature (T), Pressure (P), Weight (W), and Level (L)
- The process may fail if any one of the following conditions occur:
 - Temperature, Pressure, and Level are high
 - Low Level with high Temperature and Weight
 - Low Level and Temperature with high Pressure
 - Low Level and Weight with high Temperature

31

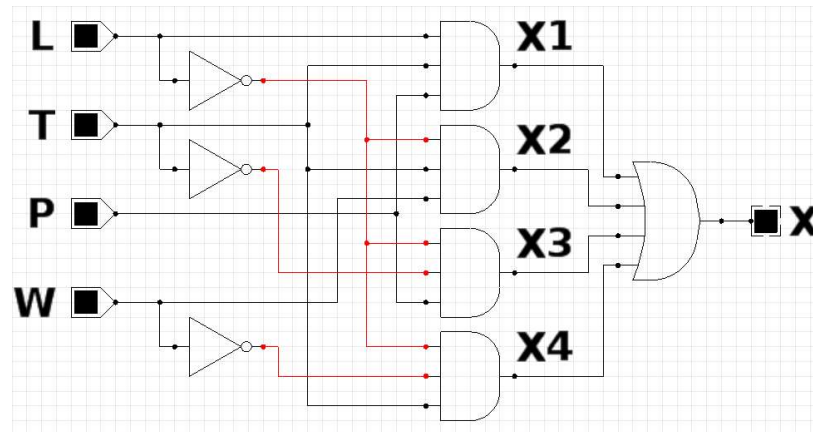
EXAMPLE (ALARM DESIGN)

- Covert the conditions to Boolean expressions
 - $X1 = LTP$
 - $X2 = L'TW$
 - $X3 = L'T'P$
 - $X4 = L'W'T$
- The alarm output becomes:
 - $X = X1 + X2 + X3 + X4$
 - $X = LTP + L'TW + L'T'P + L'W'T$

1. Temperature, Pressure, and Level are high
2. Low Level with high Temperature and Weight
3. Low Level and Temperature with high Pressure
4. Low Level and Weight with high Temperature

32

EXAMPLE (ALARM DESIGN)



33

EXAMPLE (ALARM DESIGN)

- Let's simplify the circuit/expression, $X = LTP + L'TW + L'T'P + L'W'T$

- Truth Table

L	P	T	W	X	L	P	T	W	X
0	0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	1	0
0	0	1	0	1	1	0	1	0	0
0	0	1	1	1	1	0	1	1	0
0	1	0	0	1	1	1	0	0	0
0	1	0	1	1	1	1	0	1	0
0	1	1	0	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1

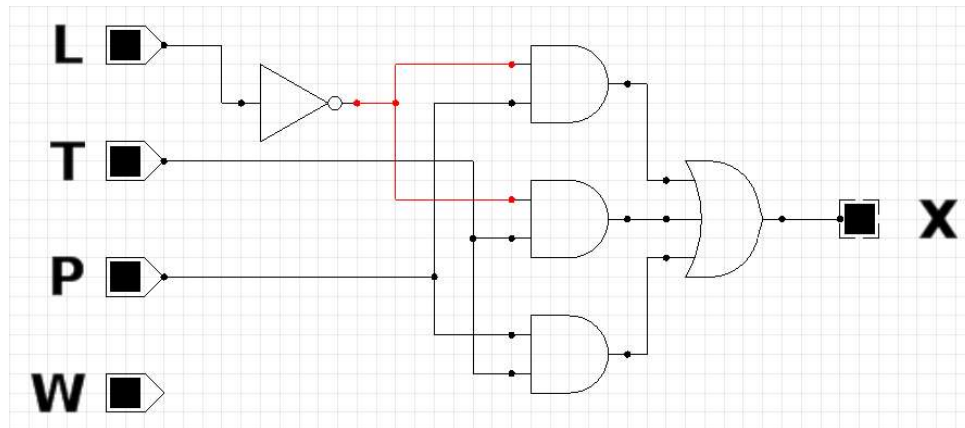
- K-map

	T'W'	T'W	TW	TW'
L'P'			1	1
L'P	1	1	1	1
LP			1	1
LP'				

- Then we have: $X = L'P + L'T + PT$

34

EXAMPLE (ALARM DESIGN)



35

ANY QUESTIONS ?

36